9 K-d Trees: Multi-dimensional Data Indexing

This chapter covers

- Efficiently indexing a 2-D (and in general k-D) dataset
- Implementing nearest neighbor search with k-d trees
- Discussing k-d trees strengths and flaws

This chapter will be structured slightly differently from our book's standard, simply because we will continue here a discussion started in chapter 8. Back there, we introduced a problem, searching multidimensional data for the nearest neighbor(s) of a generic point (possibly not in the dataset itself); in chapter 9, we introduce k-d trees, a data structure specifically invented to solve this problem.

In this chapter, we follow up on those topics, and so we won't introduce a new problem, but pick up the "closest hub" example from chapter 8, and show a different option to solve it, using an evolution of k-d trees, SS-trees[1].